Optimization Methodologies: Approximation Algorithms, Heuristics, and Metaheuristics

Presentation Outline: Ω

$\{\omega_1$: Approximation Algorithms (Bounded Theoretical Concepts)

ω_2: Heuristics (Functional Methodologies & Local Search)

ω_3: Metaheuristics (High-Level Conceptual Frameworks)

ω_4: Comparative Analysis (Synthesis Matrix)

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Section 1

Approximation Algorithms

Approximation Algorithms

- To achieve practical polynomial time $O(n^k)$, exact optimality must be relaxed to a rigorously bounded proximity.

Formal Definitions: Approximation Frameworks

Approximation Algorithm

- A polynomial-time algorithm for an NP-hard optimization problem that yields a feasible solution with a provable bound relative to the optimum.

Approximation Ratio (ρ)

- Minimization:
  $\rho \geq 1 \Rightarrow \rho = \max_{\text{instances}} \frac{\text{Algorithm Cost}}{\text{Optimal Cost}}$
- Maximization:
  $\rho \geq 1 \Rightarrow \rho = \max_{\text{instances}} \frac{\text{Optimal Profit}}{\text{Algorithm Profit}}$

PTAS (Polynomial-Time Approximation Scheme)

- Family of algorithms parameterized by $\epsilon > 0$. Returns a solution within $(1 + \epsilon)$ of optimum. Runs in polynomial time for fixed ϵ (e.g., $O(n^{1/\epsilon})$).

FPTAS (Fully PTAS)

- Stronger formulation. Running time is polynomial in both the input size n and $1/\epsilon$ (e.g., $O(n^3/\epsilon^2)$).

Numerical Example: Metric TSP Approximation

Example 1: 2-Approximation (MST Based)

Problem: Metric TSP on $G=(V,E)$. Objective: Min-cost cycle.

Input: $V=\{A,B,C,D\}$ on unit square. Sides=1, Diagonals=$\sqrt{2}$.

Algorithm: 1. MST T: Edges (A,B),(B,C),(C,D). Cost=3.

2. Tour from $2\times T$: $A\to B\to C\to D\to A$ (shortcut $D\to A$).

Result: Alg Cost=4. OPT Cost=4. Ratio=1. (Theoretical bound $\rho\leq 2$).

Example 2: Christofides 1.5-Approximation

Input: $V=\{0,\ldots,6\}$. Wheel graph. $w(0,i)=1$. $w(i,i+1)=1$ (cycle 1..6).

Algorithm: 1. MST T: Star with center 0. Cost=6.

2. Odd vertices $O=\{1,2,3,4,5,6\}$.

3. Min Matching M on O: $\{(1,2),(3,4),(5,6)\}$. Cost=3.

4. Tour from $T\cup M$: $0\to 1\to 2\to 3\to 4\to 5\to 6\to 0$ (with shortcuts).

Result: Alg Cost=7. OPT Cost=6 (cycle 1-..-6-1). Ratio = $\frac{7}{6}\approx 1.17$. (Theoretical bound $\rho\leq 1.5$).

Section 2

Heuristics

Definition of Heuristic Method

A heuristic is a problem-solving approach that employs a practical method not guaranteed to be optimal, perfect, or rational, but which is nevertheless sufficient for reaching an immediate, short-term goal or approximation. It involves sacrificing provable optimality guarantees to secure rapid, scalable, and computationally feasible solutions, especially in unbounded or complex search spaces where finding an exact solution is impractical.

Formal Definitions: Heuristic Classifications

Constructive Heuristics

Definition: Builds a feasible solution strictly incrementally from an empty initial state.

Formulation: $S_0 = \emptyset \rightarrow S_{k+1} = S_k \cup \{e_i\}$, stopping when S_{final} is completely feasible.

Conceptual Example: Building a tower from a pile of blocks one by one until it reaches a certain height or no more blocks fit.

Search-Based Heuristics

Definition: Iteratively modifies an existing complete feasible solution to locate a local optimum.

Neighborhood Structure $N(S)$: A defined set of nearby solutions reachable via a single local transformation operator from current state S.

Local Optimum: A state S^* where $\forall S' \in N(S^*), f(S^*) \leq f(S')$ (for minimization).

Numerical Example: Constructive Heuristic (Knapsack)

Problem Statement:

Objective: $\max \sum_i v_i x_i$ subject to $\sum_i w_i x_i \leq W$.

Constraint: Capacity $W = 50$.

Component State (Greedy Ratio Evaluation):

Items Vector $I = [i_1, \quad i_2, \quad i_3]$,

Value Vector $V = [60, \quad 100, \quad 120]$,

Weight Vector $W_{item} = [10, \quad 20, \quad 30]$,

Score Ratio $(V/W) = [6, \quad 5, \quad 4]$.

Final Constructive State:

Selection Sequence: $S_0 = \emptyset \rightarrow S_1 = \{i_1\} \rightarrow S_2 = \{i_1, i_2\}$.

Result: Knapsack load = $30 \leq 50$. Total Value = 160.

Section 3

Metaheuristics

Metaheuristics: A high-level strategy governing subsidiary heuristics to escape local optima via diversification and intensification.

- Introducing stochastic perturbations and high-level control strategies to continuously balance vast global exploration with acute local exploitation.

Formal Definitions: Metaheuristic Frameworks

General Framework

Definition: A high-level strategy governing subsidiary heuristics to escape local optima via diversification and intensification.

Simulated Annealing (SA)

Mechanics: Mimics physical thermodynamics. Probabilistically accepts worsening moves to escape local minima.
Acceptance Probability Axiom: $P(accept) = e^{-\Delta f/T}$, where T is the continuously decreasing system temperature.

Genetic Algorithms (GA)

Mechanics: Maintains a population matrix of candidate solutions, applying biologically-inspired transformations.
Operators: Selection (fitness-proportionate), Crossover (recombination), Mutation (stochastic variance).

Ant Colony Optimization (ACO)

Mechanics: Inspired by ant foraging behavior. Uses artificial pheromone trails and heuristic information to construct solutions.
Key Elements: Pheromone update (evaporation & deposition), probabilistic path selection.

Particle Swarm Optimization (PSO)

Mechanics: Simulates social behavior of bird flocks or fish schools. Particles move through search space influenced by their own best known position and the swarm's global best.
Update Rule: Velocity and position vectors updated iteratively.

Tabu Search (TS)

Mechanics: Local search method that uses a memory structure (tabu list) to avoid revisiting recent solutions and cycling.
Features: Tabu tenure, aspiration criteria to override tabu status.

Numerical Example: GA Selection Components

1. Problem Definition

Objective: Determine the selection probability for each individual in a population based on their fitness scores.

Input (Evaluation States): Population Vector $S = [s_1, s_2, s_3, s_4, s_5, s_6]$
Fitness Vector $F = [12, 4, 16, 8, 36, 24]$

2. Algorithm (How it's done)

Step 1: Calculate Total Fitness. Total Fitness Sum: $\sum F = 12 + 4 + 16 + 8 + 36 + 24 = 100$.

Step 2: Calculate Selection Probability. Formula: $P(x_i) = F(x_i) \;/\; \sum F_{.} = 12/\,100 = 0.12$

3. Answer (Result)

Selection Probability Vector P_{sel} = [12/100, 4/100, 16/100, 8/100, 36/10,24/100] = [0.12, 0.04, 0.16, 0.08, 0.36, 0.24].

Interpretation (Proof): Highest fitness s_5 (36) secures the highest probability (36% selection chance).

Section 4

Comparative Analysis

Methodological Overview

- **Exact Algorithms:** Guarantee optimal solutions but often require exponential computational time, making them impractical for large-scale problems (NP-Hard).
- **Approximation Algorithms:** Provide near-optimal solutions within provable bounds in polynomial time, balancing quality and speed.
- **Heuristics:** Use problem-specific rules of thumb to find good solutions quickly in practice, but lack theoretical guarantees on solution quality.
- **Metaheuristics:** High-level, stochastic frameworks designed to explore vast search spaces and escape local optima, adapting to complex problems.

Diagnostic Matrix: Methodological Methodological Trade-Offs

Method	Solution Quality	Time	Scalability	Theoretical Guarantee
Exact Algorithms	Optimal	Exponential (NP-Hard)	Weak	Strongest
Approximation Algorithms	Near-optimal with bound	Polynomial (very fast)	Good	Provable Ratio (ρ)
Heuristics	Good in practice	Polynomial (moderate)	Very good	(none, weak)
Metaheuristics	Often very good in practice	Moderate to high	Very good	(none, weak)